

Team 4 – Sentinel Fusion

Work Allocation Among Team Members

Objective: Clearly define the responsibilities assigned to each team member so that every major component of the Sentinel Fusion project is owned, developed, tested, and integrated collaboratively throughout the project lifecycle.

Ashwin Manoj – Frontend & UI

Primary Responsibilities

- Set up Next.js project and frontend architecture.
- Develop authentication screens and responsive dashboard.
- Implement Evidence Upload, Case Management, Timeline View, Relationship Graph, Risk Dashboard and Report Viewer.
- Integrate frontend with backend APIs.
- Ensure responsive UI, accessibility, testing and UI documentation.

Technology Stack: Next.js, React, Tailwind CSS, shadcn/ui, React Flow, Cytoscape.js, Apache ECharts

Expected Deliverables: Complete responsive frontend with integrated APIs.

Abhinand K – Backend & Database

Primary Responsibilities

- Develop FastAPI backend and REST APIs.
- Design PostgreSQL schema and Neo4j graph model.
- Implement authentication, file upload and Redis integration.
- Generate OpenAPI documentation.
- Support API testing and backend optimization.

Technology Stack: FastAPI, PostgreSQL, Neo4j, Redis, SQLAlchemy

Expected Deliverables: Backend services, database and API documentation.

Joyal Blesson – AI & Intelligence

Primary Responsibilities

- Develop OCR, metadata extraction and Whisper transcription.
- Implement image understanding and deepfake detection.
- Build risk scoring, timeline reconstruction and relationship extraction.
- Create LangGraph workflows and AI report generation APIs.
- Evaluate AI models and improve accuracy.

Technology Stack: Python, LangGraph, GPT-5.5 API, PaddleOCR, Whisper, OpenCV, ExifTool, Qdrant

Expected Deliverables: AI pipelines, AI APIs and model evaluation.

Rosemary Rejimon – DevOps, Security & Integration

Primary Responsibilities

- Set up Docker environment and CI/CD pipelines.
- Manage deployment, monitoring, logging and object storage.
- Configure RBAC, audit logs and security.
- Perform performance testing and final system integration.
- Maintain technical documentation and deployment guides.

Technology Stack: Docker, GitHub Actions, MinIO/S3, Prometheus, Grafana, Loki, Sentry

Expected Deliverables: Production deployment, infrastructure and security configuration.

Shared Responsibilities

- Sprint planning and task coordination.
- Code reviews and GitHub pull requests.
- Bug fixing, testing and quality assurance.
- Project documentation.
- Integration support and final presentation/demo.
- Follow branch workflow: Feature → Develop → Main.

Project Lifecycle Contribution

During Sprint 1, each member establishes their assigned subsystem. Sprint 2 focuses on core feature development. Sprint 3 emphasizes integration, testing, and bug fixing. Sprint 4 includes final polishing, documentation, deployment, and project demonstration. Although responsibilities are divided, all members collaborate during integration to ensure a secure, scalable, and fully functional AI-powered digital forensics platform.